

You need a HD63C09P (exactly this reference):



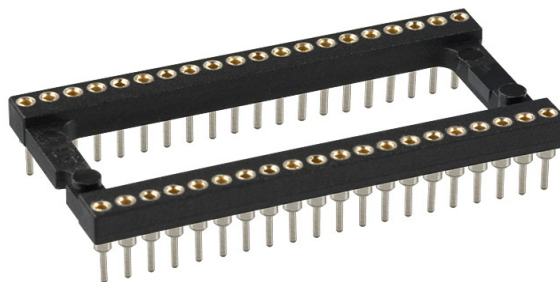
A CD4040 (or any compatible CMOS 4040):



A 16Mhz crystal:



A 40 pin IC socket:



A crystal socket (Optional, see <http://www.qrpkits.com/buildertip03.html>).

Wire as follows:

- Unsolder the 6809 CPU.
- Solder the 40 pin socket in its place.
- Unsolder the 8Mhz crystal.
- Solder the 16Mhz crystal in its place (or alternatively, the crystal socket, to be able to revert to the original CPU).
- Remove pin 34 of the 6309 (bend it until it breaks).
- Use 2 sided duct tape to glue the 4040 backwards on the top of the 6309.
- Solder pin 1 of the 6309 to pin 8 & 11 of the 4040 (GND).
- Solder pin 7 of the 6309 to pin 16 of the 4040 (5V).
- Solder pin 35 of the 6309 to pin 10 of the 4040.
- Solder a loose wire to pin 9 of the 4040, the other end will go into the hole of pin 34.

